$$\partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right) - \frac{\partial\mathcal{L}}{\partial\phi} = 0 \quad \vee \quad \frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) = 0$$

$$\underbrace{\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) = 0}_{(i)} \qquad \underbrace{\frac{\partial\mathcal{L}}{\partial\phi^*} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)}\right) = 0}_{(ii)}$$

$$\mathcal{L} = \frac{\Box\phi^2}{2} - \frac{m^2\phi^2}{2} = 0 \Rightarrow \left(\frac{\partial_t^2}{2} - \frac{\partial_i^2}{2}\right)\phi^2 - \frac{m^2}{2}\phi^2 = 0$$

$$(\partial_t\phi^*)(\partial^t\phi) - (\partial_i\phi^*)(\partial^i\phi) - m^2|\phi|^2$$

$$= (\partial_t\phi^*)(\partial^t\phi) - (\partial_i\phi^*)(\partial^i\phi) - m^2(\phi^*)(\phi) = 0$$

(ii) ye bakarsak; $\underbrace{\frac{\partial\mathcal{L}}{\partial\phi^*} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)}\right) = 0}_{(ii)}$

$$\left\{\frac{\partial}{\partial\phi^*}, \frac{\partial}{\partial(\partial_\mu\phi^*)}\right\}$$

$$\Rightarrow (\partial_\mu\phi^*)(\partial^\mu\phi) - m^2(\phi^*)(\phi) = 0$$

$$\frac{\partial}{\partial\phi^*} = -m^2\phi \quad , \quad \frac{\partial}{\partial(\partial_\mu\phi^*)} = \partial^\mu\phi$$

$$\Rightarrow \underbrace{\frac{\partial\mathcal{L}}{\partial\phi^*} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)}\right)}_{(ii)} = 0 \Rightarrow -m^2\phi - \partial_\mu(\partial^\mu\phi) = 0$$

$\Rightarrow \partial_\mu\partial^\mu\phi + m^2\phi = 0 \qquad \partial_\mu\partial^\mu = \partial_t\partial^t - \nabla^2$ Lorentz op'dür.

(ii) $(\Box + m^2)\phi = 0$

(i) $(\Box + m^2)\phi^* = 0$

$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$ $\quad \phi_1$ ve ϕ_2 skaler reel alan

$$\mathcal{L} = \frac{1}{2}\left[(\partial_\mu\phi_1)^2 + (\partial_\mu\phi_2)^2\right] - \frac{1}{2}m^2\left[\phi_1^2 + \phi_2^2\right]$$

Noether Akımının ispatı

$$\phi \to \phi' =: \phi + \delta\phi \qquad \phi^* \to \phi^{*\prime} =: \phi^* + \delta\phi^*$$

$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi) \Rightarrow \delta(\partial_\mu\phi) = \partial_\mu(\delta\phi)$ sıradeğiştirilebilir.

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \underbrace{\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi)}_{?}$$

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi) \Rightarrow \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\cdot\delta\phi\right) = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\cdot\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi)$$

$$? = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu(\delta\phi) = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\cdot\delta\phi\right) - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\cdot\delta\phi$$

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\cdot\delta\phi\right) - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\cdot\delta\phi$$

$$\Rightarrow \delta\mathcal{L} = \left\{\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right)\right\}\delta\phi + \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\cdot\delta\phi\right)$$

$$\delta \mathcal{L} = \Big\{ \underbrace{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \Big)}_{0} \Big\} \delta\phi + \partial_\mu \Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \cdot \delta\phi \Big)$$

$$\delta \mathcal{L} = \Big(\underbrace{\frac{\partial \mathcal{L}}{\partial \phi^*} - \partial_\mu \Big(\frac{\partial \mathcal{L}}{\partial_\mu \phi^*} \Big)}_{0} \Big) \delta\phi^* + \partial_\mu \Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \cdot \delta\phi^* \Big)$$

E-L denk $= 0$ idi

$$\Rightarrow \delta \mathcal{L} = \partial_\mu \Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \cdot \delta\phi \Big) + \partial_\mu \Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \cdot \delta\phi^* \Big)$$

$$\Rightarrow \delta \mathcal{L} = \partial_\mu \Big(\underbrace{\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \delta\phi^*}_{J^\mu} \Big) = 0$$

$\delta L = 0$ idi

$\partial_\mu J^\mu = 0$ Noether Teoremi ispatlandı//

$\phi = e^{i\alpha}\phi \qquad \phi^* = e^{-i\alpha}\phi^*$

$\delta\phi = i\alpha\,\phi \qquad \delta\phi^* = -i\alpha\,\phi^*$

$$(\partial_\mu \phi^*)(\partial^\mu \phi) - m^2 (\phi^*)(\phi) = 0$$

$$\frac{\partial \mathcal{L}}{\partial \phi^*} = -m^2 \phi \quad , \quad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} = \partial^\mu \phi$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = -m^2 \phi^* \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi^*$$

$$\delta \mathcal{L} = \partial_\mu \Big(\underbrace{\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \delta\phi^*}_{J^\mu} \Big) = 0$$

$$J^\mu = \partial^\mu \phi^* \, i\alpha \phi - i\alpha \phi^* \partial^\mu \phi$$

$J^\mu = i\alpha \big(\phi (\partial^\mu \phi^*) - \phi^* (\partial^\mu \phi) \big)$ $\qquad \alpha$ sbt sayı faz

$$J^\mu = i \big[\phi (\partial^\mu \phi^*) - \phi^* (\partial^\mu \phi) \big]$$

yükü oluşturmak için

$Q = \int d^3x \, \underbrace{J^0(x)}_{J^\mu(x)}$ $\qquad \partial^0 \phi = ? \quad \partial^0 \phi^* = ?$ $\qquad \partial^0 \phi = \frac{\partial \phi}{\partial t} = \dot{\phi}$

$$\partial^0 \phi^* = \frac{\partial \phi^*}{\partial t} = \dot{\phi}^*$$

$$J^\mu = i (\phi \dot{\phi}^* - \phi^* \dot{\phi})$$

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \big(a\, e^{-i\omega t + ikx} + b_k e^{+i\omega t - ikx} \big)$$

$$\phi^*(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \big(a_k^+ e^{+i\omega t - ikx} + b_k^+ e^{-i\omega t + ikx}$$

$\omega = \sqrt{|k|^2 + m^2}$

$E^2 = p^2 + m^2$

$\omega^2 = p^2 + m^2$

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} (a_k e^{-iwt+ikx} + b_k e^{+iwt-ikx}) \quad ---(1)$$

$$J^\mu = i(\phi \dot{\phi}^* - \phi^* \dot{\phi})$$
$$(1)\,(4) \qquad (2)\,(3)$$

$$\phi^*(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} (a_k^+ e^{+iwt-ikx} + b_k^+ e^{-iwt+ikx} \quad ---(2)$$

$$\dot{\phi}(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} (-iw)(a_k e^{-iwt+ikx}) + (iw)(b_k e^{iwt-ikx}) \quad ---(3)$$

$$\dot{\phi}^*(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} (iw)(a^+ e^{iwt-ikx}) - (iw)(b_k^+ e^{-iwt+ikx}) \quad ---(4)$$

$$J^\mu = i((1)(4) - (2)(3))$$

$$(1)(4) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} (a_k e^{-iwt+ikx} + b_k e^{+iwt-ikx}) \cdot \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} (iw)(a_k^+ e^{iwt-ikx}) - (iw)(b_k^+ e^{-iwt+ikx})$$

$$\int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} \Big\{ (iw)(a_k a_k^+) - (iw)(a_k b_k^+) e^{-2iwt+2ikx} + (iw)(b_k a_k^+) e^{2iwt-2ikx} - (iw)(b_k b_k^+) \Big\}$$

$$(2)(3) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} (a_k^+ e^{iwt-ikx} + b_k^+ e^{-iwt+ikx})(-iw)(a_k e^{-iwt+ikx}) + (iw)(b_k e^{iwt-ikx})$$

$$\Rightarrow \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} (-iw)(a_k^+ a_k) + (iw)(a_k^+ b_k) e^{2iwt-2ikx} + (-iw)(b_k^+ a_k e^{-2iwt+2ikx}) + (iw)(b_k^+ b_k)$$

$$i((1)(4) - (2)(3))$$

$$i\left\{ \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2w_k}} \Big\{ (iw)(a_k a_k^+) - \cancel{(iw)(a_k b_k^+) e^{-2iwt+2ikx}} + \cancel{(iw)(b_k a_k^+) e^{2iwt-2ikx}} - (iw)(b_k b_k^+) - \Big[(-iw)(a_k^+ a_k) + \cancel{(iw)(a_k^+ b_k) e^{2iwt-2ikx}} + \cancel{(-iw)(b_k^+ a_k e^{-2iwt+2ikx})} + (iw)(b_k^+ b_k) \Big] \right\}$$

$$= i\int \frac{d^3k}{(2\pi)^3} \frac{1}{2w_k} (iw) \Big[a_k a_k^+ + a_k^+ a_k - b_k b_k^+ - b_k^+ b_k \Big]$$

$$= -\int \frac{d^3k}{(2\pi)^3} \frac{1}{\cancel{2}} \cdot \cancel{2} \Big[a_k a_k^+ - b_k^+ b_k \Big]$$

$$\int \frac{d^3k}{(2\pi)^3} \Big[a_k a_k^+ - b_k^+ b_k \Big]$$

$$\int \frac{d^3k}{(2\pi)^3} \Big[a_k a_k^+ \Big] \Rightarrow \langle p' | p \rangle = \sqrt{2w_{p'}} \sqrt{2w_p} \langle 0 | a_p a_p^+ | 0 \rangle \qquad w_{p'} = w_p$$

$$\Rightarrow \langle p' | p \rangle = 2w_p \langle 0 | a_p \cdot a_p^+ | 0 \rangle = 2w_p \delta(p - p')$$

tüm ifadede; $(2\pi)^3 (2w_p) \delta(p - p')$

$$|p\rangle = \sqrt{2w_p}\, a_p^+ |0\rangle$$

$$Q = -\sqrt{2\omega_p}\int \frac{d^3k}{(2\pi)^3}[a_k^+ a_k \underbrace{- b_k^+ b_k}_{0}] a_p^+ |0\rangle$$

$$\Rightarrow -\sqrt{2\omega_p}\int \frac{d^3k}{(2\pi)^3}[a_k^+ a_k] a_p^+ |0\rangle$$

$$-\sqrt{2\omega_p}\int \frac{d^3k}{(2\pi)^3}[a_k^+ a_k a_p^+ |0\rangle] = Q|p\rangle$$

$$a_k^+ a_k a_p^+ |0\rangle = \quad [a_k, a_p^+] = a_k a_p^+ - a_p^+ a_k$$

$$a_k^+ [a_k, a_p^+] = a_k^+ (a_k a_p^+ - a_p^+ a_k)$$

$$= a_k^+ a_k a_p^+ - a_k^+ a_p^+ a_k$$

$$+ a_k^+ a_p^+ a_k$$

$$= a_k^+ a_k a_p |0\rangle = a_k^+ [a_k, a_p^+] + a_k^+ a_p^+ a_k$$

$$= a_k^+ a_k a_p |0\rangle = a_k^+ \{\underbrace{[a_k, a_p^+]}_{(2\pi)^3 (2\omega_k) \delta^3(k-p)} + a_p^+ a_k\} |0\rangle$$

$$Q|p\rangle = \sqrt{2\omega_p}\int \frac{d^3k \cancel{(2\pi)^3} \cancel{2\omega_k}}{\cancel{(2\pi)^3} \cancel{2\omega_k}} a_k^+ \delta^3(k-p) |0\rangle$$

$$Q|p\rangle = \sqrt{2\omega_p}\int d^3k \, a_k^+ \delta^3(k-p) |0\rangle = |p\rangle = \sqrt{2\omega_p}\, a_p^+ |0\rangle //$$

$$Q|p\rangle = |p\rangle$$

b^+ içinde $-\sqrt{2\omega_p}\, b_p^+ |0\rangle //$

$$-Q = q[N_a - N_b] \Rightarrow Q = q(-N_a + N_b)$$